

SAMAR FATHALAH

Software Engineer | Full Stack Engineer

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Summary

Full-stack engineer with **2+ years** building production systems in **JavaScript, Node.js, and Python**. Shipped merchant-facing fulfillment features serving **250+ merchants and 700+ daily orders** at an e-commerce startup, and internal data platforms at **Valeo Schalter und Sensoren GmbH**. M.Sc. Information Technology, University of Stuttgart.

Experience

Working Student - SW Engineer | Valeo Schalter und Sensoren GmbH - Bietigheim-Bissingen, Germany | 01/2025 - 06/2026

- Built and containerized **Flask and FastAPI REST** services with **Docker** to access and deliver **autonomous-driving test data** from **MySQL** databases, including sensor calibration data, and vehicle metadata, forming a core piece of internal simulation infrastructure.
- Developed user/client dashboards using **Grafana** to **visualize and analyze** sensor data, e.g., RADAR, LiDAR, and vehicle information from autonomous-driving test environments, enabling engineers to triage test runs without manual log inspection.
- Automated **hardware integration** using **Python/C-based** tooling, cutting environment setup from 4 hours to 20 minutes for driver installation and system configuration.

Full Stack Developer | Khazenly - Cairo, Egypt | 01/2023 - 09/2024

- Built merchant-facing features for a **self-service fulfillment platform** serving **250+ merchants**, enabling automated order fulfillment, real-time order tracking, and self-service workflow configuration without operations-team involvement.
- Integrated **60 e-commerce storefronts (Shopify, WooCommerce, etc.)** with third-party delivery and tracking APIs in **JavaScript**, enabling end-to-end order streaming across the fulfillment chain.
- Scaled automated order fulfillment flows between merchants, customers, and logistics partners using **AWS (Lambda, S3, SQS)**, improving delivery reliability and handling more than **700 orders/day**.

Teaching Assistant | The German University in Cairo - Cairo, Egypt | 08/2022 - 01/2023

- Designed course content, technical projects, and assessment materials for the **"Introduction to Media Engineering"** course for third-year students in the **Media Engineering and Technology** major.

Skills & Languages

- Languages:** English (C2), German (A2, actively improving).
- Programming Languages:** Python, JavaScript, TypeScript, Java, C/C++, C#, Go.
- Frontend:** React.js, Angular, Flutter, HTML, CSS.
- Backend:** Node.js, Express.js, Flask, FastAPI, REST APIs, WebSockets.
- Databases:** MySQL, MongoDB, PostgreSQL.
- Cloud & DevOps:** AWS (Lambda, S3, EC2, SQS, CloudWatch), Docker, Git, CI/CD.
- Methodologies:** Agile, Scrum.
- Machine Learning:** PyTorch, OpenCV, GANs, Diffusion Models.

Education

M.Sc. Information Technology (Software and Hardware Engineering) | University of Stuttgart (2024 - 2026) | GPA (2.00)

B.Sc. Media Engineering & Technology (Computer Science) | The German University in Cairo (2017- 2022) | GPA (2.2)

Projects

GUC Administration Staff Portal | **MERN Stack**

Led a team of 5 to build a **full-stack web application** used by **50+ academic staff** to manage course scheduling, room allocation, leave requests, and attendance sign-in/out, replacing a manual administrative process. Designed and implemented the **REST API and backend server** end-to-end, with role-based access control for course coordinators and teaching assistants, and a notification system triggered on administrative changes. Ran the project in **Agile/Scrum**, owning API design, sprint planning, and backlog management.

LogiSolve | **Mobile Application**

Developed a cross-platform logical riddle **quiz app** using **Flutter, GO & MongoDB** featuring user progress tracking, offline functionality, and leaderboards.

Master's Thesis & Research Project | Generative AI for Autonomous Driving & Computer Vision

Built diffusion- and GAN-based data augmentation pipelines for autonomous driving and semantic segmentation, reducing **collision rate by 39%** and achieving a best-in-class **FID of 98.01**.

Bachelor Thesis | **Automatic Music Transcription (Audio to Sheet Music)**

Developed a machine learning model using **CNN** and **signal processing techniques** to convert musical songs/notes into sheet music with an average precision of **75%**.